

Document Layout Analysis

Multi-column layouts are common in academic papers and magazines. A naive parser that groups text purely by vertical position will incorrectly merge the left and right columns into a single line. The correct approach is to first detect column boundaries by analysing the horizontal distribution of text, then read each column top to bottom.

Once columns are identified, the reading order becomes unambiguous. Each column is treated as an independent flow of blocks. This preserves the logical structure of the document and produces Markdown that mirrors how a human would read the page, left column first and right column second.